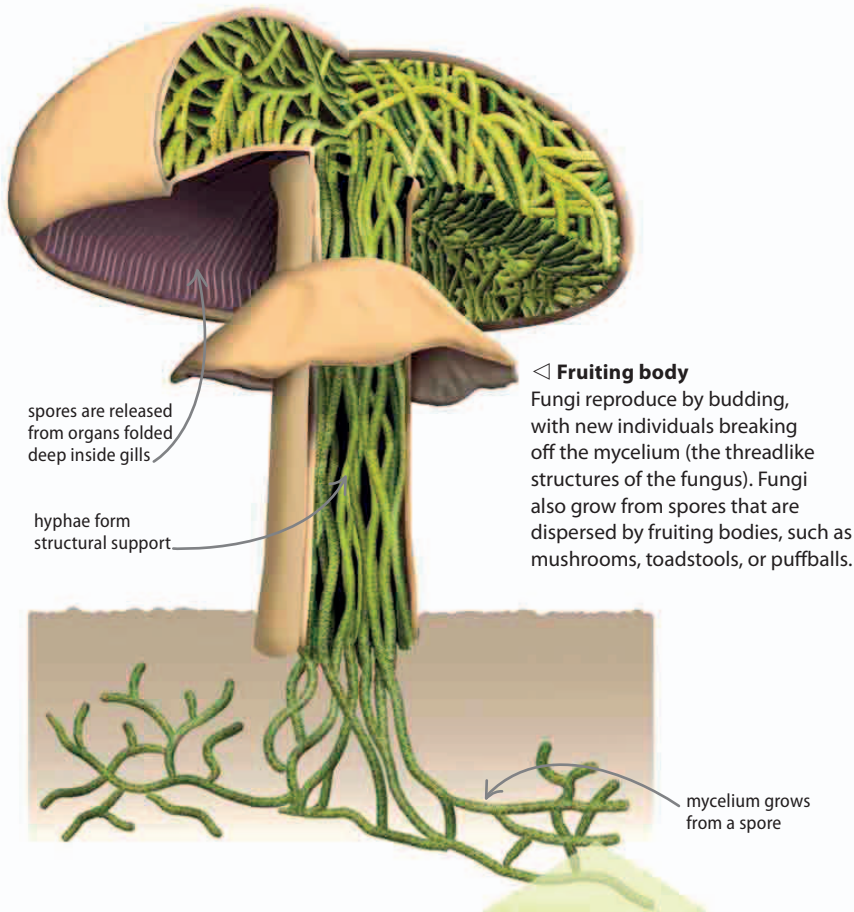


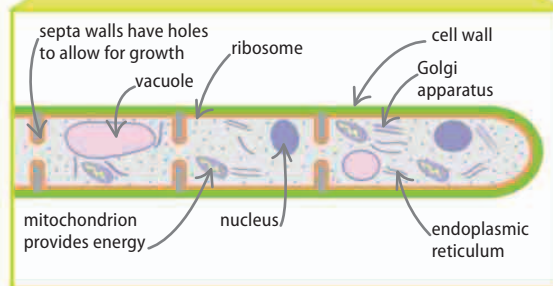
Fungi

The fungal kingdom includes mushrooms, molds, and yeasts. They are saprophytic organisms, which means they grow over a food source and secrete enzymes that digest it externally. Their cells are eukaryotic, with a nucleus and organelles like those of plants and animals. The cells are held inside a rigid cell wall made largely of chitin, the same material that crab shells and beetle wings are made of.



▷ Hypha

The main part of a fungus is called the mycelium. This is made up of many strands called hyphae, which are long tubes of cells that extend over food sources. Yeast are single-celled fungi and do not develop hyphae.

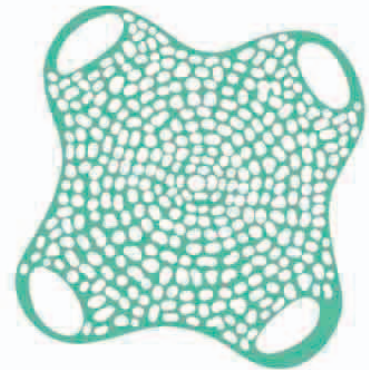


Protists

This kingdom includes a wide variety of single-celled organisms. There are at least 30 different phyla and it is likely that at least some of them evolved separately from each other. The protist cell is very diverse, and can resemble that of an animal, plant, or fungus. Some species, such as *Euglena*, photosynthesize with chloroplasts, but also feed like animals.

▽ Diatom

These single-celled algae live in sunlit waters. They have an ornate cell wall made from silica. In the right conditions, diatoms produce thick blooms in the water. The silica skeletons of dead diatoms are one of the ingredients in clay.



▽ Ciliate

Not every protist is motile (able to move). An amoeba alters the shape of its cell so its contents flow in one direction. Flagellates are powered by tail-like flagella, while ciliates (below) waft hairlike extensions called cilia (singular: cilium) to push themselves along.

